

Weight Distribution Optimization via Interior Point Methods

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Problem Setup

Let $\mathcal{L}(\Theta)$ be the loss function.

Θ be the set of parameters of the neural network.

$\Theta \sim p_{\theta}(\Theta)$; Weights follow a distribution. (Assumption).

Objective:- Minimize the “Expected Loss” over weights.

$$\therefore \min \int_{-\infty}^{\infty} p_{\theta}(\Theta) \cdot \mathcal{L}(\Theta) \cdot d\Theta$$

Constraints:- Probability axioms

$$\int_{-\infty}^{\infty} p_{\theta}(\Theta) \cdot d\Theta = 1 \quad [\text{Normalization}]$$

$$p_{\theta}(\Theta) \geq 0 \quad \forall \Theta \quad [\text{PDF must be } \geq 0]$$

Constrained Optimization & Log-Barrier

Constrained Optimization Problem:-

$$\begin{aligned} \min \quad & \int_{-\infty}^{\infty} p_{\theta}(\Theta) \cdot \mathcal{L}(\Theta) \cdot d\Theta \\ \text{s.t.} \quad & \int_{-\infty}^{\infty} p_{\theta}(\Theta) \cdot d\Theta = 1 \\ & p_{\theta}(\Theta) \geq 0 \quad \forall \Theta \end{aligned}$$

Adding Log-Barrier term

$$\begin{aligned} \min \quad & \int_{-\infty}^{\infty} p_{\theta}(\Theta) \mathcal{L}(\Theta) \cdot d\Theta - \mu \int_{-\infty}^{\infty} \log p_{\theta}(\Theta) \cdot d\Theta \\ \text{s.t.} \quad & \int_{-\infty}^{\infty} p_{\theta}(\Theta) \cdot d\Theta = 1 \end{aligned}$$

No non-negativity constraints

The Lagrangian & Derivative

Lagrangian.

$$L = \int_{-\infty}^{\infty} p_{\theta}(\Theta) \mathcal{L}(\Theta) d\Theta - \mu \int_{-\infty}^{\infty} \log p_{\theta}(\Theta) d\Theta \\ + \lambda \left[\int_{-\infty}^{\infty} p_{\theta}(\Theta) d\Theta - 1 \right]$$

$$L = -\lambda + \int_{-\infty}^{\infty} [p_{\theta}(\Theta) \mathcal{L}(\Theta) - \mu \log p_{\theta}(\Theta) + \lambda p_{\theta}(\Theta)] \cdot d\Theta$$

Derivative of the Lagrangian

$$\frac{dL}{d\Theta} = \mathcal{L}(\Theta) - \frac{\mu}{p_{\theta}(\Theta)} + \lambda$$

KKT Conditions

KKT conditions

1. **Stationarity:-** $\mathcal{L}(\Theta) + \lambda - \frac{\mu}{p_{\theta}(\Theta)} = 0$
2. **Feasibility:-** $\int_{-\infty}^{\infty} p_{\theta}(\Theta) \cdot d\Theta = 1$; $q_{\theta}(\Theta) \geq 0 \forall \Theta$
3. **Complementary Slackness:-** $p_{\theta}(\Theta) \cdot q_{\theta}(\Theta) = \mu$

Stationarity & Limits

From stationarity,

$$p_{\theta}^*(\Theta) = \frac{\mu}{\mathcal{L}(\Theta) + \lambda}$$

$$\text{Also, } \int_{-\infty}^{\infty} p_{\theta}(\Theta) \cdot d\Theta = 1 \quad \therefore \mu = \frac{1}{\int_{-\infty}^{\infty} \frac{d\Theta}{\mathcal{L}(\Theta) + \lambda}}$$

$$\text{As } \mu \rightarrow 0, \int_{-\infty}^{\infty} \frac{d\Theta}{\mathcal{L}(\Theta) + \lambda} \rightarrow \infty$$

$$\therefore \mathcal{L}(\Theta) + \lambda \rightarrow 0$$

$$\boxed{\therefore \lambda = -\min \mathcal{L}(\Theta)} \quad \text{At optimum}$$

Stationarity Analysis

Therefore

$$\mathcal{L}(\Theta) - \frac{\mu(\Theta)}{p_{\theta}(\Theta)} + \lambda(\Theta) = 0 \text{ (Stationarity).}$$

$$\mu(\Theta) = p_{\theta}(\Theta)\{\mathcal{L}(\Theta) + \lambda(\Theta)\} \quad \underline{= 0 \text{ (Optimal)}}$$

$$\mu'(\Theta) = p'_{\theta}(\Theta)\{\mathcal{L}(\Theta) + \lambda(\Theta)\} + \{\mathcal{L}'(\Theta) + \lambda'(\Theta)\}p_{\theta}(\Theta)$$

At optimality, $\{\mathcal{L}(\Theta) + \lambda(\Theta)\} \rightarrow 0$

$$\therefore \mu'(\Theta) = \{\mathcal{L}'(\Theta) + \lambda'(\Theta)\}p_{\theta}(\Theta).$$

$$\boxed{\therefore \mu'(\Theta) = \mathcal{L}'(\Theta)p_{\theta}(\Theta).}$$

Updating the Parameters

Push $\mu \rightarrow 0$, with,

$$p_{\theta}(\Theta)\{\mathcal{L}(\Theta) + \lambda(\Theta)\} = \mu(\Theta)$$

$$\mu'(\Theta) = \mathcal{L}'(\Theta) \cdot p_{\theta}(\Theta)$$

At optimum, $\Theta = \Theta^*$, $\mathcal{L}(\Theta^*) = -\lambda(\Theta)$; $\mu(\Theta^*) = 0$.